

SiamUF: SiamCar based small UAV tracker using dense U-shape deep features in near infrared videos

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ABSTRACT

Existing feature extraction methods demonstrate poor performance when tracking a small infrared Unmanned Aerial Vehicle (UAV) against complex backgrounds. For a small target, even a slight positional deviation can cause a significant drop in the corresponding intersection over union. To tackle these challenges, a Siamese tracker leveraging a robust feature extraction model is developed to track a small infrared target. Specifically, with the base model SiamCAR, a U-shaped deep feature extraction model is incorporated to capture robust features that refer to a small target. Next, the side window filter reduces the negative effects of edges on tracking small targets. To address the problem of target deviation, the Wasserstein distance is embedded in the regression head and the template updating mechanism is constructed, improving the tracking stability. Furthermore, the proposed model is trained using a dataset consisting of public near-infrared UAV videos, which aids in acquiring precise network parameters. The experimental results demonstrate that the proposed SiamUF algorithm offers superior performance and enhanced stability compared to both correlation filtering trackers and popular Siamese trackers, achieving an overall precision of 83.1%. The code can be accessed at <https://github.com/Moriarty288/SiamUF>, thus benefiting the optical information processing community.

1. Introduction

The Unmanned Aerial Vehicle (UAV) becomes increasingly integrated into daily life, evolving into highly accessible technology [1]. It is crucial to monitor small UAVs effectively to ensure public safety. Using infrared data for tracking offers advantages such as enhanced stealth and reliable performance in various weather scenarios. However, the reduced resolution of infrared images introduces multiple difficulties in monitoring small UAVs in city landscapes [2–4]. Moreover, tracking can be easily disrupted due to environmental thermal cross-interference and disturbances from small objects that resemble each other.

Recently, infrared small target tracking methods in complex background include mainly the Discriminative Correlation Filter (DCF) [5–9]. The DCF model acquires a rich set of candidate samples and balances speed and accuracy to some extent by solving a ridge regression model [10]. The Kernelized Correlation Filter (KCF) achieves high speed through kernel techniques and performs well in feature extraction. Qian et al. [6] integrated a background update strategy with the KCF model to adapt to tracking small infrared targets. However, manual parameter

settings make it sensitive to complex backgrounds. The Spatio-Temporal Regularized Correlation Filter (STRCF) [11], which incorporates spatial and temporal information, is insensitive to boundary effects and also obtains a high tracking speed. The guided image filter has been incorporated into the KCF tracker [6] to achieve more effective samples. To predict the target location, the Kalman filter is integrated with STRCF [8]. Moreover, enhancing the robustness of the tracking model is achieved by integrating the detection model with an adaptive search range factor. Nevertheless, these algorithms for tracking small targets often struggle to extract effective features from such targets.

Based on the aforementioned information, the use of Deep Learning (DL) [12–15] in small target tracking is essential. Popular DL methods include Fully Convolutional Networks (FCN) [16], Siamese networks [17] and transformer [18,19]. For the first DL tracking [14], the dataset used for training lacks diversity. Although the FCN method improves the richness of the training data with a convolutional network, its real-time performance is not high. Siamese networks hold a significant position in DL tracking. The Siamese Instance Search Tracker (SINT) [17] tends to transform the challenge of tracking tar-

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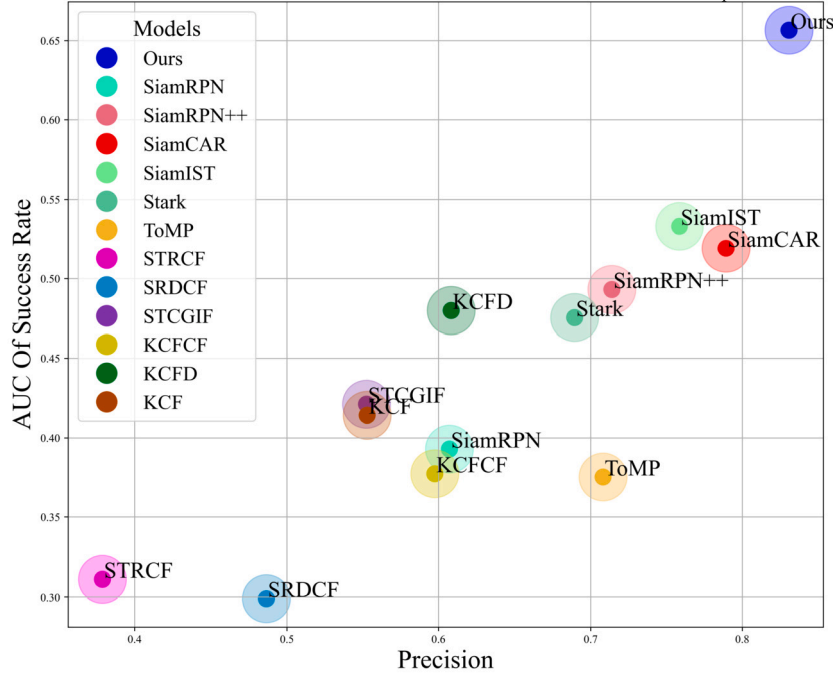


Fig. 1. The scatter plot of the tracking performance. The closer the points are to the top right corner, the better the model performance.

gets into a patch matching problem. The Fully Convolutional Siamese Network (SiamFC) [20], which uses a twin network model, obtains the target position referring to similarity maps from the two branches. In addition, the high-performance Siamese Region Proposal Network (SiamRPN) [21] and SiamRPN++ [22] take advantage of AlexNet [23] and ResNet [24]. However, the lack of texture information that refers to small targets makes it difficult to extract robust features. Additionally, pooling operations of DL tracking algorithms may lead to the loss of targets' detail. The Siamese Infrared Small Target Tracker (SiamIST) [25], which recalibrates the SiamRPN model, achieves the perception of infrared image patterns. With transfer learning techniques, the overfitting issue that refers to a small UAV is solved to some extent. In the paper Siamese Fully Convolutional Classification and Regression for Visual Tracking (SiamCAR) [26], SiamCAR tends to design an anchor-free framework, which increases its speed. However, the Intersection over Union (IoU) value can be lower if the target drifts to the background, which negatively affects the tracking capability.

Consequently, this paper develops an algorithm based on the Siamese [26] network to track a small target. Building on the lightweight SiamCAR model, we present a Densely Nested U-shaped Feature (DNUF) model to extract deep features for small infrared targets. An improved regression loss is given to address the issue of target deviation. It is worth mentioning that a publicly available small target video set is used for training. This method makes full use of cascaded channel and spatial attention models, enhancing the performance of the feature extraction model. Furthermore, the Side Window Filter (SWF) [27] is used to minimize the impact of the background edge. In addition, a Template Update (TU) [28] mechanism is used to adjust to the evolving appearance of the target over time. Experimental results demonstrate that the SiamUF algorithm (SiamCAR enhanced with U-shaped features) is capable of accurately tracking small UAVs even in challenging backgrounds. The primary contributions are described below.

- A lightweight Siamese technique with enhanced loss function is developed to track a small UAV target in a publicly available near-infrared video dataset. The SiamUF tracker is characterized by high tracking precision, which helps regulate small urban drones.
- The nested U-shaped network (Unet) can capture effective target details, enhancing the ability to track a faint target.

- The SWF mitigates the negative impact of edges in infrared images, thus accentuating the feature of small targets. In addition, the template update mechanism enables the tracking model to adjust to environmental variations.
- The normalized Wasserstein distance is incorporated into the loss calculation as a new similarity metric to mitigate the issue of overly aggressive IoU loss changes for small objects.

The main subjects covered in this paper are outlined below. Section 1 contains the introduction. Section 2 reviews the related work. Section 3 introduces the proposed tracking algorithm. Section 4 presents the experimental findings and their analysis. Finally, Section 5 wraps up the paper.

2. Related work

In this section, the motivation of the proposed algorithm that takes advantage of the Siamese network and the Unet is given below.

In recent years, Siamese trackers have led the field in tracking performance. The computational cost of SINT [17] and SINT++ [29] is high due to its global search strategy. Consequently, SiamFC [20] can directly compute the similarity score map and locate the target in a single step, eliminating the need for a global search. However, SiamFC struggles to handle complex deformations because it lacks a robust localization mechanism. Therefore, SiamRPN [21], which applies region proposal networks, simultaneously performs classification and regression of the bounding box. Furthermore, SiamRPN++ [22] embeds a deeper residual network into feature extraction, allowing the tracker to achieve better performance in challenging tracking scenarios that involve complex backgrounds and occlusions. Nevertheless, the prediction of the target position relies on predefined anchor boxes, potentially restricting the model's adaptability in some cases. To address this issue, Siamese Box Adaptive Network (SiamBAN) [30] classifies the target directly and regresses its bounding box within a unified network. Moreover, SiamCAR [26] uses one or more prediction heads to directly make predictions, eliminating the negative impact of anchors. And Siamese Attention Network (SiamAttn) [31] introduces an attention mechanism to focus on key regions, which significantly improves tracking precision in complex backgrounds. Moreover, Siamese Network with Spatio-

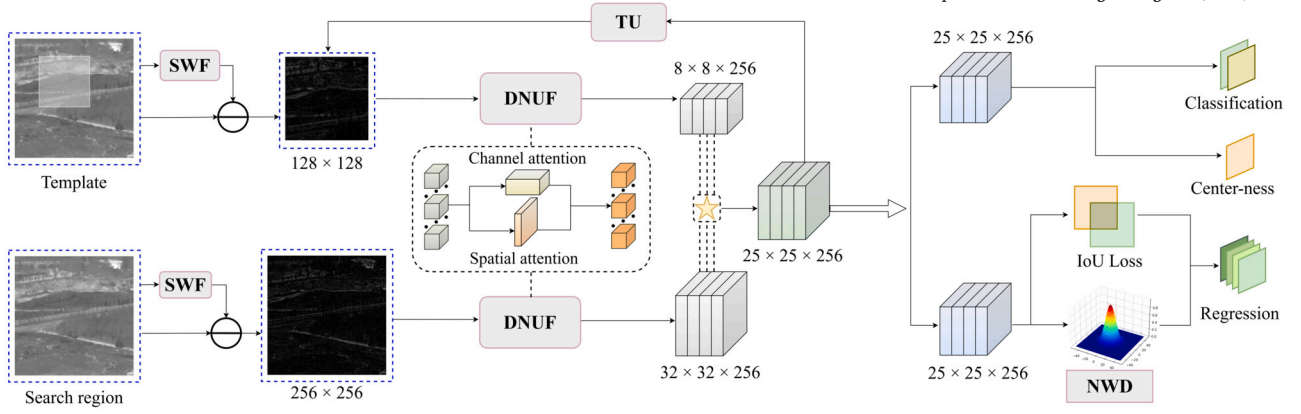


Fig. 2. The flowchart of the proposed SiamUF algorithm.

temporal Awareness (SiamST) [32] fully leverages the spatio-temporal state of the target to achieve more robust feature matching results.

The above Siamese trackers is suitable for tracking area targets in visual videos. To our knowledge, it is ineffective in the field of small target tracking due to insufficient information and the complex background [33]. Specifically, smaller targets take up fewer pixels in the images, leading to a lack of sufficient detail [34]. Furthermore, classical trackers without training exhibit weak generalization capabilities for infrared small targets with blurred edges. The tracking model can be affected by background clutter, especially if the small target is moving quickly [35]. Based on it, there is an urgent need to design a more effective feature extraction method to improve the performance of infrared small target tracking methods.

The classic feature extraction model, which contains the color and histogram, cannot adapt well to the complex background [36]. The early network, which is relatively shallow, struggles to capture intricate details in complex scenarios. To address this issue, VGG [37] and ResNet [24] are introduced into models such as SiamRPN++ and SiamBAN. Furthermore, lightweight networks such as MobileNet [38] have been employed in resource-constrained scenarios, achieving a balance between efficiency and precision. However, these feature extraction models still face limitations in processing multi-scale information and preserving spatial details. Consequently, we use Unet, which has seen extensive application in various domains, including 3D image processing [39], image enhancement [40], and small target detection, to improve small target tracking. Its U-shaped architecture, with skip connections that combine high-resolution features from early layers with deeper representations, excels in tasks requiring precise localization, such as handling complex backgrounds or small objects.

3. Proposed SiamUF tracker

3.1. Overview

Inspired by the achievements of Siamese trackers such as SiamRPN [21], SiamCAR [26], and SiamIST [25], we created SiamUF, which employs a deep U-shaped feature extraction model, illustrated in Fig. 2. Unlike traditional Siamese trackers, the proposed SiamUF possesses a strong feature that takes into account background clutter. Crucially, the Normalized Wasserstein Distance (NWD) is proposed to mitigate the problem in which even minor positional shifts can result in a substantial decrease in the IoU value. In the following sections, we will thoroughly introduce each module of SiamUF.

3.2. Siamese tracker

The anchor-free Siamese primarily involves two components, the feature extraction and the classification-regression models, as depicted in

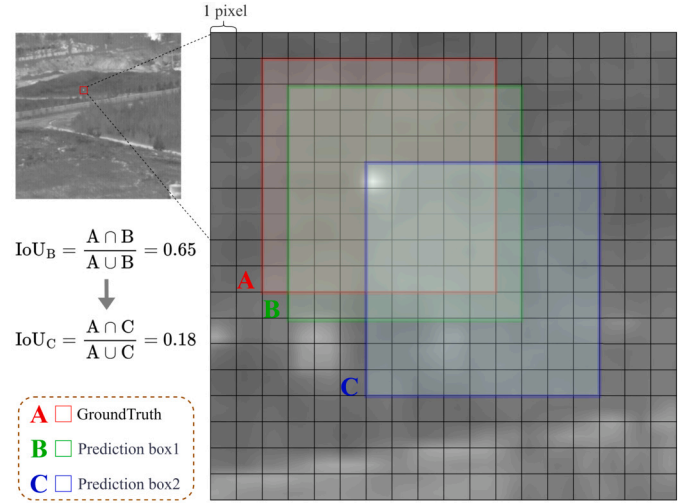


Fig. 3. A challenge of IoU Loss in Small Object Tracking. Due to the pixel-level scale of localization, even slight positional shifts can cause a significant drop in the IoU.

Fig. 2. Given the tracking template Z and the search area X , the related response map $\mathcal{R}^*$ is defined as

$$\mathcal{R}^* = f^{1 \times 1}(\Phi(Z) \star \Phi(X)) \quad (1)$$

where $f^{1 \times 1}$, $\star$ and Φ denote the 1×1 convolution operation, the cross-correlation operation, and the response function.

Subsequently, the total loss L is comprised of the classification loss L_{cl} , the regression loss L_{re} , and the centerness loss L_{ce} , and is defined as

$$L = L_{cl} + \lambda_1 L_{re} + \lambda_2 L_{ce} \quad (2)$$

where λ_1 and λ_2 are penalty factors referring to all losses. Generally, L_{re} is tasked with forecasting the location of the bounding box. When handling a small object, even a slight change in its position can significantly lower the IoU value, which could cause the model to believe that the detection box is misaligned with the ground truth.

As a result, we improve the regression loss using the NWD combined with the traditional IoU loss (Fig. 3). In the NWD method, the predicted bounding boxes as well as the ground-truth bounding boxes are depicted as the Gaussian distributions [41]. This method allows for a reliable evaluation of similarity, even when there is little overlap between the two boxes. The overall regression loss is given by

$$L_{re} = \frac{1}{\sum Q} \sum Q \times (\beta L_W + (1 - \beta) L_I) \quad (3)$$

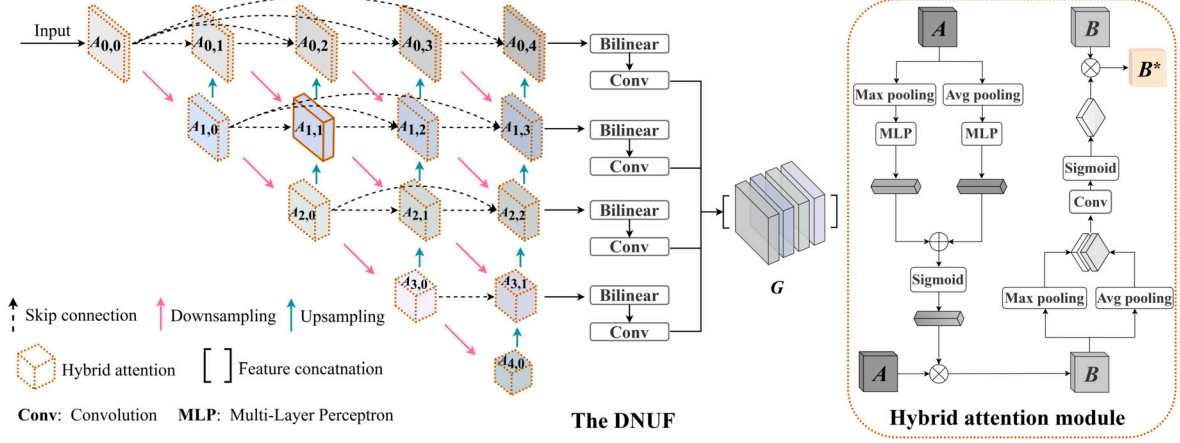


Fig. 4. The developed DNUF diagram.

where the weight β relate to the contribution of the NWD loss L_W and the IoU loss [42] L_I . If a point falls outside the boundary of the initial bounding box, then Q is set to 0. Conversely, if it is inside, Q is set to 1.

In (3), the NWD loss L_W that refers to the predicted center $c_p = (x_p, y_p)$ and ground truth $c_g = (x_g, y_g)$ can be represented by

$$L_W = 1 - \exp \left(- \frac{\sqrt{\|c_p - c_g\|_2^2 + \|(w_p, h_p) - (w_g, h_g)\|_F^2}}{\gamma} \right) \quad (4)$$

where (w_p, h_p) represent the width and height of the predicted bounding box, while (w_g, h_g) denote the width and height of the ground truth box. The symbols $\exp$, $\|\cdot\|_2$, and $\|\cdot\|_F$ represent the exponential function, the L2 norm, and the Frobenius norm, respectively. In addition, the normalization factor γ is set to 2.5 in this paper. This leads to minimal loss if the predicted and ground truth boxes are near each other but results in higher loss as the distance between them increases.

3.3. Developed DNUF

To extract more effective target features, a superior feature extraction network inspired by Unet is designed. As shown in Fig. 4, each node acquires attributes from neighboring nodes or those within the same layer, allowing the extraction of richer semantic details across an extensive receptive field. The output $A_{i,j}$ of each node can be formulated as

$$A_{i,j} = \begin{cases} \mathbf{P}_m(\mathbf{F}(A_{i-1,j})) & j = 0 \\ [\mathbf{F}[A_{i,k}]_{k=0}^{j-1}, \mathbf{P}_{\max}(\mathbf{F}(A_{i+1,j-1})), \mathbf{U}(\mathbf{F}(A_{i-1,j}))] & j > 0 \end{cases} \quad (5)$$

where $\mathbf{P}_m$, $\mathbf{F}$, $\mathbf{U}$ and $[\cdot]$ represent the max pooling, multiple cascaded convolution, the upsampling, and the concatenation of feature maps.

To effectively reduce semantic gaps in hyperspectral images, a Hybrid Attention Module (HAM) is used that incorporates channel attention and spatial attentions. The feature map A at the current node is sequentially passed through a 1D channel attention $\mathbf{M}_c$ followed by a 2D spatial attention $\mathbf{M}_s$ as described in the following.

$$B = \mathbf{M}_c(A) \otimes A \quad (6)$$

$$B^* = \mathbf{M}_s(B) \otimes B \quad (7)$$

where $\otimes$ denotes element-wise multiplication. B^* is the resulting feature map, representing the feature map B that has been improved by channel attention. $\mathbf{M}_c$ and $\mathbf{M}_s$ are given by

$$\mathbf{M}_c(A) = \sigma[\mathbf{M}(\mathbf{P}_m(A)), \mathbf{M}(\mathbf{P}_a(A))] \quad (8)$$

$$\mathbf{M}_s(B) = \sigma[f^7(\mathbf{P}_m(B)), \mathbf{P}_a(B)] \quad (9)$$

where σ , $\mathbf{M}$, $\mathbf{P}_a$ and f^7 denote the sigmoid function, the multi-layer perceptron, the average pooling, and the 7×7 convolution.

By applying two successive enhancement operations to the input feature map, the feature representation capability is effectively strengthened, improving the precision of feature selection. In addition, solely $A_{3,1}$, $A_{2,2}$, $A_{1,3}$, and $A_{0,4}$ are chosen to derive the multilayer feature, which encapsulates both shallow spatial details and deep semantic content. The method for calculating the robust global feature map G can be explained as

$$G = [f^1(\mathbf{I}(A_{0,4})), f^1(\mathbf{I}(A_{1,3})), f^1(\mathbf{I}(A_{2,2})), f^1(\mathbf{I}(A_{3,1}))] \quad (10)$$

where $\mathbf{I}$ represents bilinear interpolation.

3.4. Side window filter

When the edges of the image partially occlude a UAV target, it greatly impacts tracking performance [43]. To address this, side window filtering is employed to reduce edge effects. SWF considers eight directions (up, down, left, right, northeast, southeast, southwest, northwest) for the filtered window. For a region x , the filtered result $x^{r,\rho,\theta}$ of SWF can be given by

$$x^{r,\rho,\theta} = \mathbf{W}(x, r, \rho, \theta) \quad (11)$$

where $\mathbf{W}$ refers to the SWF function, with r being the radius of the filtered window. A value of $\rho = 0$ signifies that the pixel is on the edge of the window, while $\rho = r$ means that it is on the corner. Furthermore, θ takes values of $0, \pi/2, \pi, 3\pi/2$, representing the four directions of the filtered window.

In addition, the ideal result $\delta(x)$ is achieved by reducing the discrepancy between the input x and the output $x^{r,\rho,\theta}$.

$$\delta(x) = \arg \min_{\forall x^{r,\rho,\theta}} \|x - x^{r,\rho,\theta}\|_2^2 \quad (12)$$

where $\|\cdot\|_2$ represents the 2-norm, and the information about the target, x , is improved by $(x - \delta(x))$.

The edge-preserving capability of the SWF model is demonstrated in infrared images. In SWF, the radius of the filter window is 1, and the process undergoes 3 iterations. As depicted in Fig. 5, the complex background is suppressed and the target retains high-frequency information, effectively distinguishing the target and the edge.

3.5. Template update

In continuous tracking, small infrared targets may undergo appearance changes due to variations in pose and illumination variation. As a

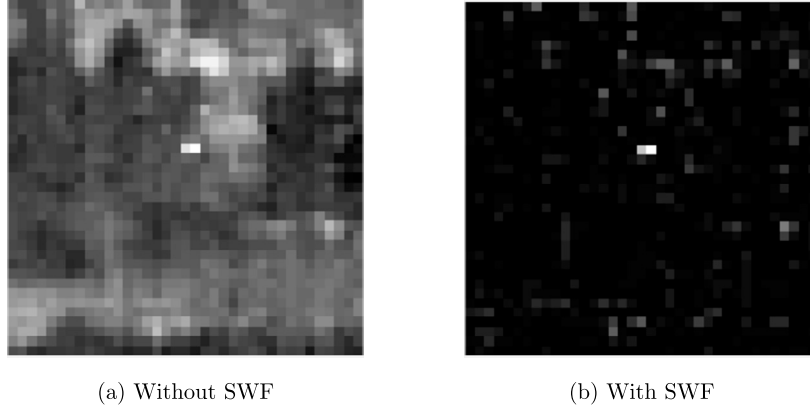


Fig. 5. The comparison results referring to SWF. (b) clearly shows that the contour boundaries of background objects surrounding the target are effectively suppressed.

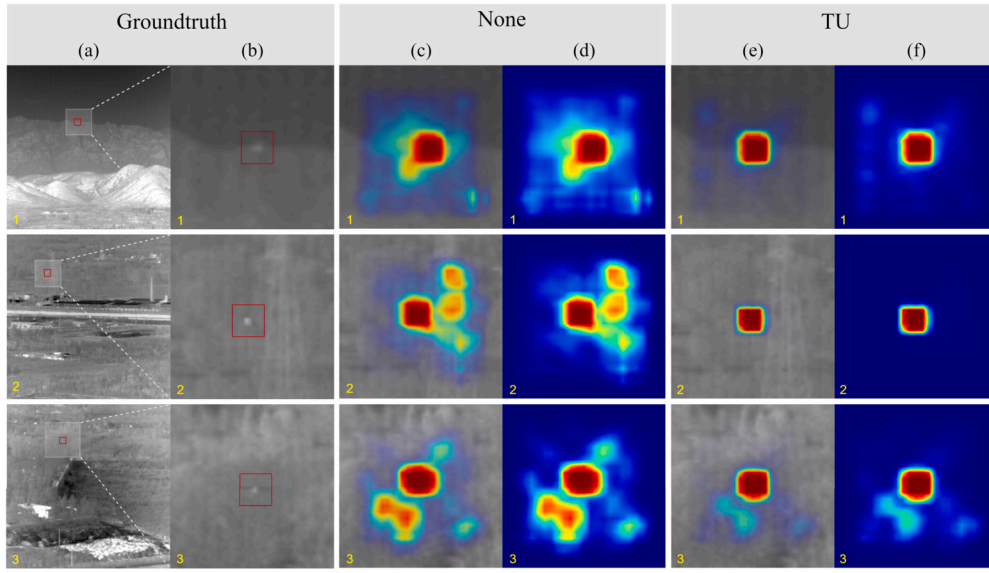


Fig. 6. Visualization of response maps in three videos (from top to down, data10, data15 and data17). Columns (a)-(f) show the original images, the magnified search regions, the original response maps (fused with background), the original response maps, the response maps with TU (fused with background) and the response maps with TU, respectively.

result, the initial template may no longer accurately depict the current appearance of the target. As the tracking duration extends, template matching can become less reliable due to background noise or other interfering elements [44].

To address these issues, a template update mechanism is incorporated into the model, as shown in Fig. 2. In particular, the current target information is combined with the preceding template to create an updated template that preserves historical information. Here, the updated template T_{t+1} is represented by

$$T_{t+1} = (1 - \alpha) \cdot T_t + \alpha \cdot I_t \quad (13)$$

where T_t and I_t denote the target template and the detection result for the t -th frame, respectively. The update coefficient α signifies the importance of current versus historical information.

By updating the template cumulatively, the tracker maintains a consistent focus on the target, which improves tracking stability. Fig. 6 shows that the response map is affected by local interference in the absence of TU. Furthermore, the updated template substantially decreases this interference.

4. Experiments and analysis

The experimental setup is described in Section 4.1, including the selected data set and experimental details. In Section 4.2 and Section 4.3, the qualitative and quantitative analyzes are performed with several existing small object tracking methods. The strengths and limitations of the proposed algorithm are discussed. Furthermore, in Section 4.4, ablation experiments are performed to demonstrate the effectiveness of each component in tracking small objects in near-infrared videos.

4.1. Experiment setup

Small target datasets from public IR sources [45,46] are utilized for both training and testing. The training dataset originates from the paper titled ‘A dataset for small infrared moving target detection under clutter background’ [45]. The chosen videos mainly display targets that are 3×3 pixels in size. In addition, this dataset also includes diverse conditions during image capture, such as varying altitudes, motion scenarios, and complex backgrounds. Additionally, the test dataset is derived from the work titled ‘A dataset for infrared image dim small aircraft target

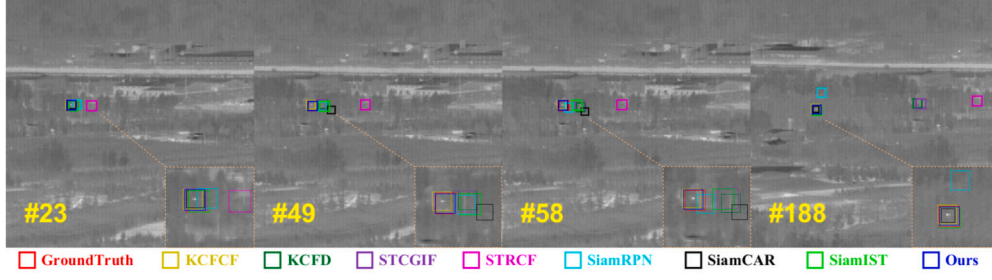


Fig. 7. Qualitative results on video 5. In the presence of complex edges and grayscale variations. (Best viewed in color display).

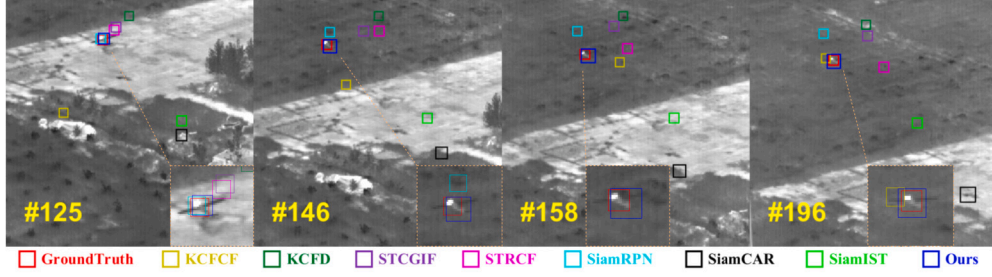


Fig. 8. Qualitative results on video 9. Situations where the background texture and color feature is similar to the target. (Best viewed in color display).

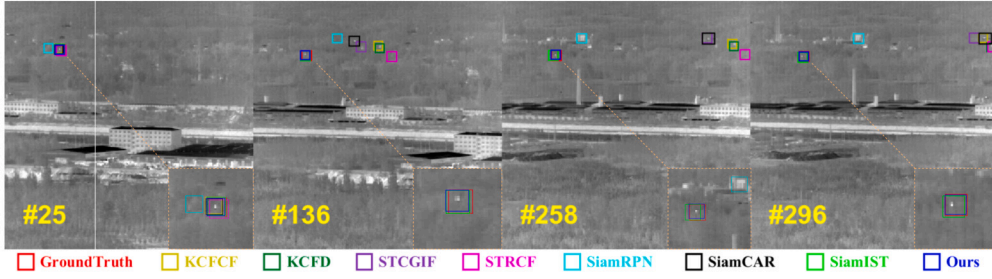


Fig. 9. Qualitative results on video 15. Situations with other objects similar to the target. (Best viewed in color display).

detection and tracking under ground/air background' [46]. It is important to mention that the initial four videos are not considered due to the presence of either oversized or multiple targets.

The suggested approach has been run with Pytorch on a GTX3090 GPU, which boasts 24GB of memory. Given the small size of the dim target, the SWF window radius is configured to 1. Throughout the training phase, we configure the batch size to 16 and conduct 20 epochs in total, using stochastic gradient descent with an initial learning rate of 0.001. During the initial 10 epochs, our primary focus is training the classification and regression networks. In the subsequent 10 epochs, we unlock the previously fixed parameters of the feature extraction model, thereby incorporating Unet into the training process. In the regression loss, $\beta = 0.2$ and $\gamma = 2.5$. In the TU, $\alpha = 0.0102$. In addition, the comparison algorithms contain KCF [10], KCFCF [6], SRDCF [47], KCFD [48], STRCF [11], STCGIF [49], Stark [50], ToMP [51], SiamRPN [21], SiamRPN++ [22], SiamCAR [26] and SiamIST [25]. To illustrate the efficacy of the proposed algorithm, Figs. 7 to 11 show the experimental results in various representative videos.

4.2. Qualitative comparison

In Fig. 7, the target in video 5 initially appears against a background of sparse vegetation. In frames #49 and #58, models such as SiamIST and the original SiamCAR shows significant drift, due to the complex edge and grayscale variation. The STRCF inaccurately identifies the contour of a background object as the target. When the background surrounding the target became stable, most models adjusted their tracking errors, evident in frame #188. However, the SiamRPN struggles to make these

corrections due to its weak feature extraction network. With the support of the SWF, the proposed SiamUF reliably and efficiently monitors the target during the full duration of the video sequence.

In Fig. 8, the target moves through a region with low grayscale values. Because the background texture and color features are similar to the target, the DCF models (KCFCF, KCFD, STCGIF, and STRCF) are almost entirely unable to track the target. Meanwhile, SiamIST and SiamCAR are also misled by background edges or noise, consistently detecting the wrong target. Despite the fact that the proposed method encounters some interference in target box regression, leading to dimensions exceeding the target's width and height, the target is accurately positioned with success.

In Fig. 9, the background includes point-like buildings very similar to the target. By frame #136, most models are misled and unable to track the correct target. The SiamCAR model consistently follows a building protrusion because it was too similar to the target. The proposed model accurately tracks the target by effectively merging shallow features with deep semantic information, fully extracting the target feature.

In Fig. 10, due to movement during filming, the video is relatively blurry and has low resolution. Under these conditions, the KCFCF model failed completely to track the target. In frame #156, the STRCF model rapidly drifts from the target and is unable to recover it. The tracking performance of SiamRPN and SiamIST is unstable, often deviating from the target. In contrast, SiamCAR and our method perform well.

In Fig. 11, the background is complex, with a lot of vegetation and buildings with varying light and dark areas. Due to the distinct features of the target compared to the surrounding background, most tracking

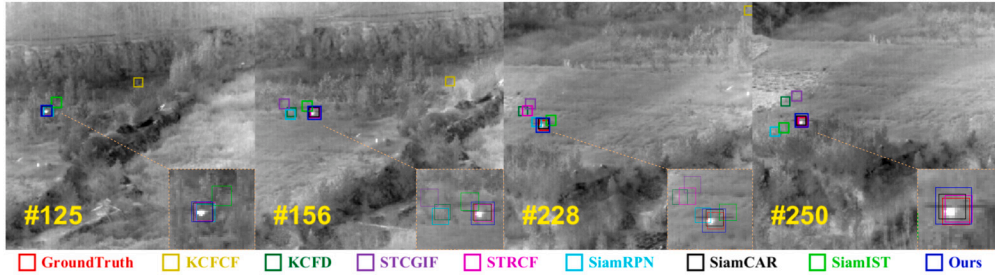


Fig. 10. Qualitative results on video 16. Scenes affected by the source of capture. (Best viewed in color display).

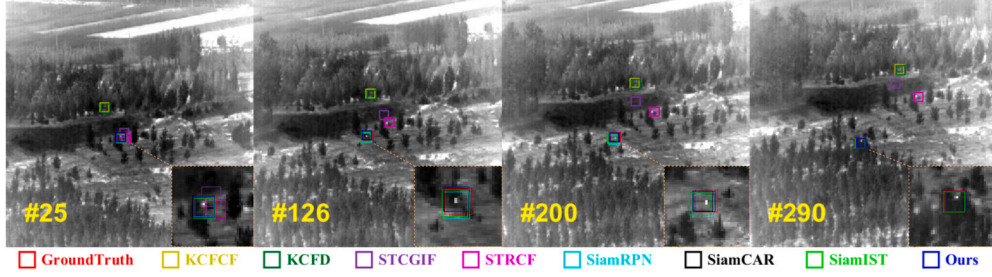


Fig. 11. Qualitative results on Video 22. Situations with complex backgrounds. (Best viewed in color display).

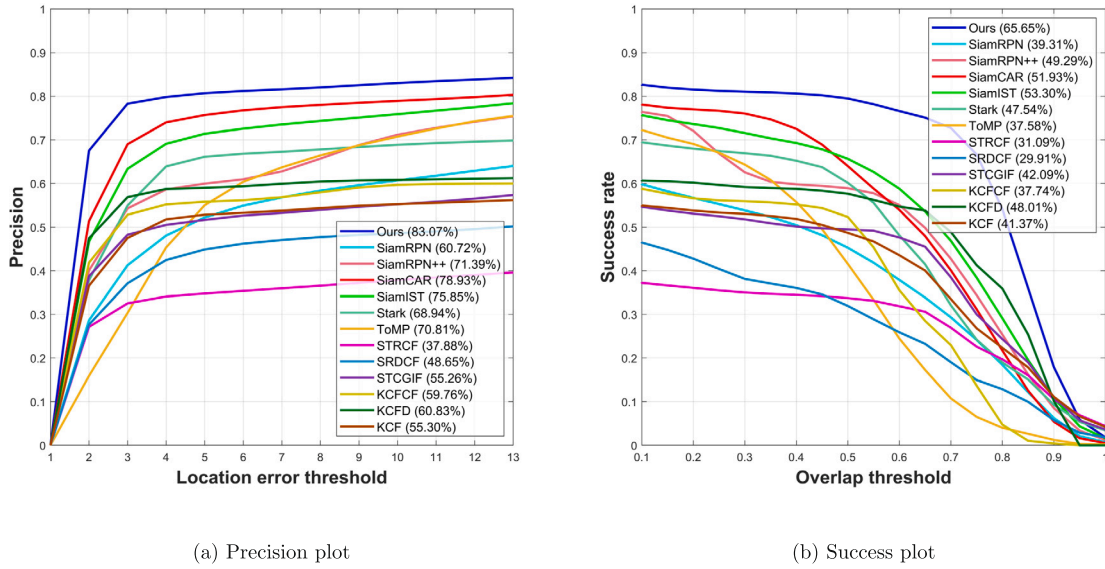


Fig. 12. The performance plots. The legends in the precision plot and the success plot display the precision and the AUC of the success rate, respectively.

models based on the Siamese network achieved good results. However, models based on the KCF method performed poorly, especially KCFCF, which has a large tracking error throughout.

4.3. Quantitative comparison

In this section, we begin by using the precision plot and the success plot [52] to assess performance. The precision is determined by calculating the ratio of frames in which the Center Location Error (CLE) d_c between the estimated bounding box and the actual reference falls below a certain threshold to the total number of frames. The success rate measures the percentage of frames where the Intersection over Union (IoU) between the predicted bounding box and the ground truth surpasses a certain threshold, relative to the overall number of frames. Furthermore, the Area Under the Curve (AUC) of the success curve represents the area located below the performance curve when plotted against the horizontal axis. Furthermore, to offer a clearer representation of overall performance, Fig. 1 presents a performance scatter plot.

As illustrated in Fig. 12, the proposed algorithm shows a 4% higher precision and significantly exceeds SiamCAR in success rate by 14%. Compared to SiamIST, there is a significant improvement in both precision and success rate, with improvements of 7% and 12%, respectively. Moreover, the average performance of DCF algorithms is notably lower compared to different DL methods.

Furthermore, Table 1 provides a comparative analysis that includes precision, AUC of the success rate curves, and Frame per Second (FpS) metrics in all infrared videos. Among DCF methods, the STRCF algorithm has a very high FpS, reaching up to 146. However, its precision is the lowest at only 0.38. The KCFD, KCFCF, and SiamRPN algorithms have comparable accuracy and also achieve high FpS. Compared to the baseline, the FpS of SiamUF is only half due to the increased computational parameters of Unet and SWF. For tracking small targets in infrared images, the proposed algorithm achieves commendable precision (0.83) and an AUC value of 0.66, with a CLE d_c of 10.

Table 1

The performance comparison (Precision refers to $d_c = 10$).

Algorithm	FpS ↑	Precision ↑	AUC ↑
STRCF [11]	146	0.3788	0.3113
SRDCF [47]	58	0.4865	0.2991
STCGIF [49]	52	0.5526	0.4212
KCF [10]	205	0.5530	0.4142
KCFCF [6]	69	0.5976	0.3774
SiamRPN [21]	65	0.6072	0.3930
KCFD [48]	70	0.6083	0.4801
Stark [50]	28	0.6894	0.4757
ToMP [51]	20	0.7081	0.3754
SiamRPN++ [22]	63	0.7139	0.4932
SiamIST [25]	31	0.7585	0.5331
SiamCAR [26]	66	0.7893	0.5192
SiamUF (Ours)	30	0.8307	0.6566

Table 2

The ablation experiment.

Baseline	SWF	DNUF	TU	NWD	AUC ↑	Precision ↑	FpS ↑
✓	-	-	-	-	0.5192	0.7893	66
✓	✓	-	-	-	0.5204	0.7981	34
✓	-	✓	-	-	0.6459	0.8150	34
✓	✓	✓	-	-	0.6467	0.8245	32
✓	-	-	✓	-	0.5221	0.7920	52
✓	-	-	-	✓	0.5411	0.7953	66
✓	✓	✓	✓	✓	0.6566	0.8307	30

SWF, DNUF, TU and NWD denotes side window filter, densely nested Unet feature, template update and normalized Wasserstein distance, respectively.

In summary, the proposed algorithm shows strong performance, which is supported by both visual results and outstanding evaluation metrics.

4.4. The ablation study

To verify the effectiveness of the proposed algorithm compared to the baseline, we incrementally added SWF and the Dense Nested U-Net Features (DNUF) module to SiamCAR and tested on the same infrared dataset.

As shown in Table 2, the baseline itself has a very good precision of 0.7893. However, Table 1 indicates that the corresponding AUC of the baseline success rate is relatively low. As shown in Table 2, both SWF and DNUF contribute to improving tracking performance. The baseline-SWF achieves an overall precision of 0.8050, which is better than the baseline algorithm (0.7893). Baseline-SWF-DNUF attains the highest precision (0.8307) and significantly increases the corresponding AUC of the success rate to 0.6566. In summary, these results demonstrate the effectiveness of each module in the proposed model.

5. Conclusion

This paper presents an improved SiamCAR algorithm tailored to track small UAV targets in near-infrared videos. The U-shaped deep feature extraction model, which takes advantage of the hybrid attention technology, adapts to the infrared information. The enhanced regression loss takes advantage of the IoU and NWD losses, which mitigates the negative impact of target migration. Moreover, choosing an appropriate radius in the SWF aids in mitigating the negative effect of background edges, and an effective template update allows the tracker to adapt to changes in the target's appearance. The superior quality of the training set within the public dataset is crucial for the generalization of the proposed tracker. Crucially, the experimental results indicate that the SiamUF algorithm achieves an overall precision of 0.83 when applied to small targets. This lays the groundwork for utilizing a widely-used deep learning model to monitor small targets.

CRedit authorship contribution statement

Kun Qian: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jian Shen:** Writing – review & editing, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation. **Shuchang Wang:** Visualization, Validation, Software, Resources, Investigation, Data curation. **Yutong Wu:** Visualization, Software, Resources, Investigation. **Guoxiang Lu:** Writing – review & editing, Visualization, Resources, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] Jiang N, Wang K, Peng X, Yu X, Wang Q, Xing J, et al. Anti-uav: a large-scale benchmark for vision-based uav tracking. *IEEE Trans Multimed* 2021;25:486–500.
- [2] Chen L, Chen X, Rao P, Guo L, Huang M. Space-based infrared aerial target detection method via interframe registration and spatial local contrast. *Opt Lasers Eng* 2022;158:107131.
- [3] Wang K, Du S, Liu C, Cao Z. Interior attention-aware network for infrared small target detection. *IEEE Trans Geosci Remote Sens* 2022;60:1–13.
- [4] Xi Y, Zhou Z, Jiang Y, Zhang L, Li Y, Wang Z, et al. Infrared moving small target detection based on spatial-temporal local contrast under slow-moving cloud background. *Infrared Phys Technol* 2023;134:104877.
- [5] Bolme DS, Beveridge JR, Draper BA, Lui YM. Visual object tracking using adaptive correlation filters. In: 2010 IEEE computer society conference on computer vision and pattern recognition. IEEE; 2010. p. 2544–50.
- [6] Qian K, Zhou H, Rong S, Wang B, Cheng K. Infrared dim-small target tracking via singular value decomposition and improved kernelized correlation filter. *Infrared Phys Technol* 2017;82:18–27.
- [7] He Z, Zhang Z, Jung C. Fast Fourier transform networks for object tracking based on correlation filter. *IEEE Access* 2018;6:6594–601.
- [8] Feng S, Hu K, Fan E, Zhao L, Wu C. Kalman filter for spatial-temporal regularized correlation filters. *IEEE Trans Image Process* 2021;30:3263–78.
- [9] Qian K, Wang J-s, Zhang S-j. Strcf: small maneuvering object tracking via improved strcf and redetection in near infrared videos. *IEEE Access* 2024.
- [10] Henriques JF, Caseiro R, Martins P, Batista J. High-speed tracking with kernelized correlation filters. *IEEE Trans Pattern Anal Mach Intell* 2014;37(3):583–96.
- [11] Li F, Tian C, Zuo W, Zhang L, Yang M-H. Learning spatial-temporal regularized correlation filters for visual tracking. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. p. 4904–13.
- [12] Zhang F, Lin S, Xiao X, Wang Y, Zhao Y. Global attention network with multiscale feature fusion for infrared small target detection. *Opt Laser Technol* 2024;168:110012.
- [13] Hao X, Luo S, Chen M, He C, Wang T, Wu H. Infrared small target detection with super-resolution and yolo. *Opt Laser Technol* 2024;177:111221.
- [14] Wang N, Yeung D-Y. Learning a deep compact image representation for visual tracking. *Adv Neural Inf Process Syst* 2013;26.
- [15] Ma C, Huang J-B, Yang X, Yang M-H. Hierarchical convolutional features for visual tracking. In: Proceedings of the IEEE international conference on computer vision; 2015. p. 3074–82.
- [16] Wang L, Ouyang W, Wang X, Lu H. Visual tracking with fully convolutional networks. In: Proceedings of the IEEE international conference on computer vision; 2015. p. 3119–27.
- [17] Tao R, Gavves E, Smeulders AW. Siamese instance search for tracking. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 1420–9.

- [18] Chen X, Yan B, Zhu J, Wang D, Yang X, Lu H. Transformer tracking. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2021. p. 8126–35.
- [19] Xie J, Zhong B, Mo Z, Zhang S, Shi L, Song S, et al. Autoregressive queries for adaptive tracking with spatio-temporal transformers. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR); 2024. p. 19300–9.
- [20] Bertinetto L, Valmadre J, Henriques JF, Vedaldi A, Torr PH. Fully-convolutional siamese networks for object tracking. In: Computer vision–ECCV 2016 workshops: Amsterdam, the Netherlands, October 8–10 and 15–16, 2016, proceedings, part II 14. Springer; 2016. p. 850–65.
- [21] Li B, Yan J, Wu W, Zhu Z, Hu X. High performance visual tracking with siamese region proposal network. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. p. 8971–80.
- [22] Li B, Wu W, Wang Q, Zhang F, Xing J, Yan J. Siamrpn++: evolution of siamese visual tracking with very deep networks. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2019. p. 4282–91.
- [23] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. *Adv Neural Inf Process Syst* 2012;25.
- [24] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 770–8.
- [25] Qian K, Zhang S-j, Ma H-y, Sun W-j. Siamist: infrared small target tracking based on an improved siamrpn. *Infrared Phys Technol* 2023;134:104920.
- [26] Guo D, Wang J, Cui Y, Wang Z, Chen S. Siamcar: Siamese fully convolutional classification and regression for visual tracking. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2020. p. 6269–77.
- [27] Yin H, Gong Y, Qiu G. Side window filtering. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2019. p. 8758–66.
- [28] Zhao F, Zhang T, Song Y, Tang M, Wang X, Wang J. Siamese regression tracking with reinforced template updating. *IEEE Trans Image Process* 2020;30:628–40.
- [29] Wang X, Li C, Luo B, Tang J. Sint++: robust visual tracking via adversarial positive instance generation. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR); 2018.
- [30] Chen Z, Zhong B, Li G, Zhang S, Ji R, Tang Z, et al. Target-aware tracking with siamese box adaptive network. *IEEE Trans Pattern Anal Mach Intell* 2022;45(4):5158–73.
- [31] Yu Y, Xiong Y, Huang W, Scott MR. Deformable siamese attention networks for visual object tracking. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2020. p. 6728–37.
- [32] Zhang H, Xing W, Yang Y, Li Y, Yuan D. Siamst: Siamese network with spatio-temporal awareness for object tracking. *Inf Sci* 2023;634:122–39.
- [33] Zhang W, Cong M, Wang L. Algorithms for optical weak small targets detection and tracking. In: International conference on neural networks and signal processing, 2003. Proceedings of the 2003, vol. 1. IEEE; 2003. p. 643–7.
- [34] Bai K, Wang Y, Yan Y, Song Q. Infrared small target tracking based on target and interference behaviors model. *Infrared Phys Technol* 2014;67:256–65.
- [35] Chen P, Zhou Y. The review of target tracking for uav. In: 2019 14th IEEE conference on industrial electronics and applications (ICIEA). IEEE; 2019. p. 1800–5.
- [36] Yucong S, Shuqing G. Traffic sign recognition based on hog feature extraction. *J Measur Eng* 2021;9(3):142–55.
- [37] Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. Preprint. arXiv:1409.1556, 2014.
- [38] Sinha D, El-Sharkawy M. Thin mobilenet: an enhanced mobilenet architecture. In: 2019 IEEE 10th annual ubiquitous computing, electronics & mobile communication conference (UEMCON). IEEE; 2019. p. 0280–5.
- [39] Çiçek Ö, Abdulkadir A, Lienkamp SS, Brox T, Ronneberger O. 3d u-net: learning dense volumetric segmentation from sparse annotation. In: Medical image computing and computer-assisted intervention–MICCAI 2016: 19th international conference, Athens, Greece, October 17–21, 2016, proceedings, Part II 19. Springer; 2016. p. 424–32.
- [40] Oktay O, Schlemper J, Folgoc LL, Lee M, Heinrich M, Misawa K, et al. Attention u-net: learning where to look for the pancreas. Preprint. arXiv:1804.03999, 2018.
- [41] Soodak RE. Two-dimensional modeling of visual receptive fields using gaussian sub-units. *Proc Natl Acad Sci* 1986;83(23):9259–63.
- [42] Yu J, Jiang Y, Wang Z, Cao Z, Huang T. Unitbox: an advanced object detection network. In: Proceedings of the 24th ACM international conference on multimedia, MM '16. New York, NY, USA: Association for Computing Machinery; 2016. p. 516–20.
- [43] Wang T, Qin R, Chen Y, Snoussi H, Choi C. A reinforcement learning approach for uav target searching and tracking. *Multimed Tools Appl* 2019;78:4347–64.
- [44] Dai K, Zhang Y, Wang D, Li J, Lu H, Yang X. High-performance long-term tracking with meta-updater. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2020. p. 6298–307.
- [45] Sun X, Guo L, Zhang W, Wang Z, Hou Y, Li Z, et al. A dataset for small infrared moving target detection under clutter background. *Chin Sci Data* 2021;5(6):8.
- [46] Hui B, Song Z, Fan H, Zhong P, Hu W, Zhang X, et al. A dataset for infrared detection and tracking of dim-small aircraft targets under ground/air background. *China Sci Data* 2020;5(3):291–302.
- [47] Danelljan M, Hager G, Shahbaz Khan F, Felsberg M. Learning spatially regularized correlation filters for visual tracking. In: Proceedings of the IEEE international conference on computer vision; 2015. p. 4310–8.
- [48] Qian K, Rong SH, Cheng KH. Anti-interference small target tracking from infrared dual waveband imagery. *Infrared Phys Technol* 2021;118:103882.
- [49] Qian K, Zhou H, Qin H, et al. Infrared dim-small target tracking based on guide filter and spatiotemporal context learning. *J Acta Photon Sin* 2015;44(9):151–6.
- [50] Yan B, Peng H, Fu J, Wang D, Lu H. Learning spatio-temporal transformer for visual tracking. In: Proceedings of the IEEE/CVF international conference on computer vision; 2021. p. 10448–57.
- [51] Mayer C, Danelljan M, Bhat G, Paul M, Paudel DP, Yu F, et al. Transforming model prediction for tracking. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2022. p. 8731–40.
- [52] Liu Q, He Z, Li X, Zheng Y. Ptb-tir: a thermal infrared pedestrian tracking benchmark. *IEEE Trans Multimed* 2019;22(3):666–75.